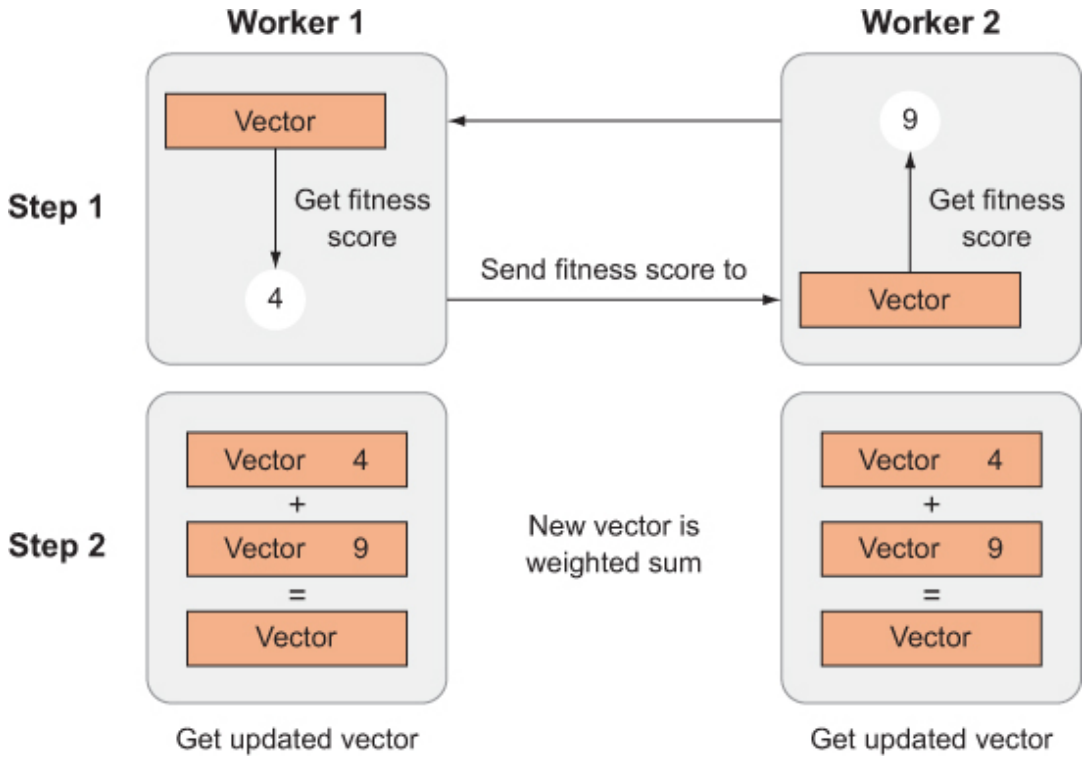


Setting the random seed allows us to consistently generate the same random numbers every time, even on different machines. If you run the code in listing 6.14, you will get the output shown, even though these numbers should be generated "randomly."

Listing 6.14. Setting the random seed

```
import numpy as np
np.random.seed(10)
np.random.rand(4)
>>> array([0.77132064, 0.02075195, 0.63364823, 0.74880388])

np.random.seed(10)
np.random.rand(4)
>> array([0.77132064, 0.02075195, 0.63364823, 0.74880388])
```

Seeding is important; it allows experiments involving random numbers to be reproduced by other researchers. If you do not supply an explicit seed, the system time or some other sort of variable number is used. If we came up with a novel RL algorithm, we would want others to be able to verify our work on their own machines. We would want the agent that another lab generated to be identical, to eliminate any source of error (and therefore doubt). That's why it's important we provide as much detail about our